

Lesson 2.1: The Concept of Uncertainty

Decision-making under uncertainty

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Introduction to uncertainty

Example (Raffle)

A man holds a hat with several raffle tickets inside it—he won't tell you how many. He tells you that if you draw a blue ticket, you win \$3—he won't tell you how many blue tickets there are. If you draw a ticket other than a blue one, you get nothing. It costs \$1 to play.

Definition (Decisions under uncertainty)

Situations in which:

- ▶ some actions have multiple possible outcomes
- ▶ probabilities unknown

Formulation

	b	$\bar{b}$
P	2	-1
R	0	0

actions play (P), refuse (R)

states draw blue (b), draw other ($\bar{b}$)

outcomes win (\$2), lose (-\$1), status quo (\$0)

- Contrast with certainty: P is not deterministic; outcome/payoff depends on b

Decision rules

	b	$\bar{b}$
P	2	-1
R	0	0

Definition (Decision rule)

A *decision rule* or *decision procedure* is a way of choosing, for each decision problem, a set of ‘preferred’ actions.

- what is a rational way of making decisions under uncertainty?